

Technical Datasheet: Strength Parameters of the Higgs Lattice

Theory: Higgs Bubble (Lattice Symmetry Theory) **Author:** Peter Kyjovsky **ORCID:** 0009-0008-3806-1964

1. Geometric Parameters

- **Structure:** Mackay Icosahedron
- **Node Count:** 55
- **Coupling Angle:** 15.5° (3D Cone)
- **Radius:** $\sim 1.7 \text{ fm}$ ($1.7 \times 10^{-15} \text{ m}$)

2. Energy Limits (Strength)

- **Total Binding Energy:** 82.67 GeV (Mean)
- **Activation Energy (1 Node):** 1.50 GeV
 - *Note:* Interactions with energy below 1.5 GeV cannot create a bubble (elastic reflection without disruption).
- **Destruction Limit (Max Load):** 147.44 GeV
 - *Note:* Exceeding this energy causes the lattice to lose structural integrity (failure to form bubbles, chaotic breakdown).

3. Mechanical Properties of the Vacuum

- **Internal Tension (Force):** $\sim 7.76 \text{ MN}$ (Million Newtons)
 - *Equivalent:* The thrust of a Falcon 9 rocket concentrated into a point smaller than a proton.
- **Stiffness:** Extremely high. The vacuum behaves as a super-hard crystal, not a fluid.

4. Conclusion

The Higgs lattice is an extremely resilient structure. Its disruption requires concentrated energy (GeV) at a precise point. However, once the activation threshold (1.5 GeV/node) is breached, a massive force ($\sim 7.76 \text{ MN}$) is released in a time shorter than light takes to cross a nucleus, explaining the destructive magnitude of the phenomenon (MET).